$[1, 2, 4, 3, 5, 6]$ n

Sub arrays. [] ✓ [1] ✓ [1, 2] ✓ [4, 3, 5] ✓ [4, 5] ✗

Subsets. [] ✓ [1] ✓ [1, 4] ✓ 2^n

$1 + (n + (n-1) + (n-2) + \dots 1)$

$1 + \frac{n(n+1)}{2}$

$[-1, 2, 3, -9, 5] \rightarrow 5$

maxSum = ~~-1~~ ~~4~~ ~~5~~ 5

currSum = ~~0~~ ~~-1~~ ~~1~~ ~~4~~ ~~-5~~ 0 ~~2~~ ~~5~~ ~~-4~~ 1

i: 0 → n-1
 current sum = 0
 j: i → n-1
 current sum += arr[j].

TC: $O(n^2)$
AS: $O(1)$

arr: [-1 2 3 -9 5]
maxTillHere: [-1 2 5 ④ ⑤]

maxTillHere[i] = max(arr[i], maxTillHere[i-1] + arr[i])

TC: $O(n)$
AS: $O(n)$

Max Circular Subarray Sum.

$[1, 2, -9, 5, 3] \rightarrow 11$

1 2 -9 5 3

$[1, 2, -9, 5, 3, 1, 2, -9, 5, 3]$

$[1, 2, 3] \rightarrow [1, 2, 3, 1, 2, 3]$

2 Possibilities.

✓ 1) Max sum is without considering the array as circular.
✓ 2) " " " with " " " "

$[-1, 2, 3, 4, -5]$
1st category.

$[2, 3, -9, 3, 4]$
2nd category.
min Sum = -9
Total sum = 3
max Sum = 3 - (-9)
 = 12

⑨ 8

Prefix Sum

$a_0, a_1, a_2, a_3, \dots a_{n-1}$

Prefix
a_0
$a_0\ a_1$
$a_0\ a_1\ a_2$
$a_0\ a_1 \dots a_{n-1}$

Suffix.
a_{n-1}
$a_{n-2}\ a_{n-1}$
$a_0 \dots a_{n-2}\ a_{n-1}$

$PS[i] = a_0 + a_1 + a_2 \dots + a_i$
$SS[i] = a_i + a_{i+1} + a_{i+2} + \dots + a_{n-1}$

arr: [1, 2, 3, 2, 5, 6]
PS: [1, 3, 6, 8, 13, 19]
SS: [19, 18, 16, 13, 11, 6]

TC = O(n)
TC = O(n)

→ [1, 3] : [1, 2, 3]
→ [5, 10] : [5, 6, 7, 8, 9, 10]
→ [6, 9] : [6, 7, 8, 9]

O/P: 6

TC: O(maxx * n)
AS: O(maxx)

freq.
1 → 1
2 → 1
3 → 1
5 → 1
→ 6 → ~~1~~ 2
7 → ~~1~~ 2
8 → ~~1~~ 2
9 → ~~1~~ 2
10 → 1

~10^{12}

0 1 2 3 4 5 ⑥ 7 8 9 10 11
| +1 | | | -1 | +1 | +1 | | | -1 | -1 | - - - -
→ 0 +1 +1 +1 0 +1 +2 +2 +2 +2 +1 0

L[i] → +1
R[i]+1 → -1

TC: O(n + maxx)
AS: O(maxx).

Binary Search.

0 1 2 3 4 5 6 7
[3, 4, 7, 9, 11, 13, 15, 20] target = ~~13~~ ~~15~~ 15

[11, 13, 15, 20]

[15, 20]

n [3, 4, 7, 9, 11, 13, 15, 20] target = 11

n/2 [11, 13, 15, 20]

n/4 [11]

1

TC: O(log(n))

mid = $\frac{(low + high)}{2}$

low = 0
high = 0

First occ.

BF: Find any occ. using BS & then hop towards left.

TC: O(n)

0 1 2 3 4 5 6 7 8
[1, 3, 5, 5, 5, 5, 67, 123, 125] x = 5

res = ~~4~~ 2

TC: O(log(n))
AS: O(1)

Sqrt.

N = 30

i = 1 - - -
2
3
4
→ 5
6

BF: TC: $O(\sqrt{n})$

[1, N]

low = 1 high = 30
mid = 15
low = 1 high = 14
mid = 7
low = 1 high = 6
mid = 3
low = 4 high = 6
mid = 5
low = 6 high = 6
mid = 6
low = 6 high = 5 → STOP

res = 3

res = 5

TC = O(log n)
AS: ~~O(n)~~ O(1).

[5.0, 5.1, 5.2 - - - - - -, 5.9] → O(1)

O(log(n) + k) → sent till kth decimal place.

$5 + \frac{⓪}{10}, 5 + \frac{①}{10}, 5 + \frac{②}{10}, 5 + \frac{③}{10} - - - , 5 + \frac{⑨}{10}$

low = 0 high = 9
mid = 4
[val = $5 + \frac{4}{10}$ = 5.4] res = 5.4
low = 5 high = 9
mid = 7
val = $5 + \frac{7}{10}$ = 5.7
low = 5, high = 6
mid = 5
val = 5.5